Calibration results
==========
Normalized Residuals

Reprojection error (cam0): mean 0.2472711207349797, median 0.19336030888472508, std: 0.22219276207976152
Gyroscope error (imu0): mean 0.202463595840656, median 0.1292403850783247, std: 0.2891305812071178
Accelerometer error (imu0): mean 0.2601905936715319, median 0.17301075664387439, std: 0.3698485567292087

Residuals

Reprojection error (cam0) [px]: mean 0.2472711207349797, median 0.1933603088847251, std: 0.22219276207976152
Gyroscope error (imu0) [rad/s]: mean 0.007994102167132889, median 0.005102946226682886, std: 0.01141607406613135
Accelerometer error (imu0) [m/s^2]: mean 0.10101845814759684, median 0.06717106730298192, std: 0.143592934785584

Transformation (cam0):

T_ci: (imu0 to cam0):
[[-0.22445104 -0.9518654   0.20874383 -0.03557569]
 [0.38626864 -0.28356571 -0.87771694 0.01879738]
 [0.89466098 -0.11637328 0.43132237 -0.01751757]
 [0.   0.   0.   1.   ]]

T_ic: (cam0 to imu0):
[[-0.22445104 0.38626864 0.89466098 0.00042645]
 [-0.9518654 -0.28356571 -0.11637328 -0.03057155]
 [0.20874383 -0.87771694 0.43132237 0.03148071]
 [0.   0.   0.   1.   ]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
0.007970614446713634

Gravity vector in target coords: [m/s^2]
[-0.19715823 -9.40868826 2.75792255]

Calibration configuration
==========

cam0

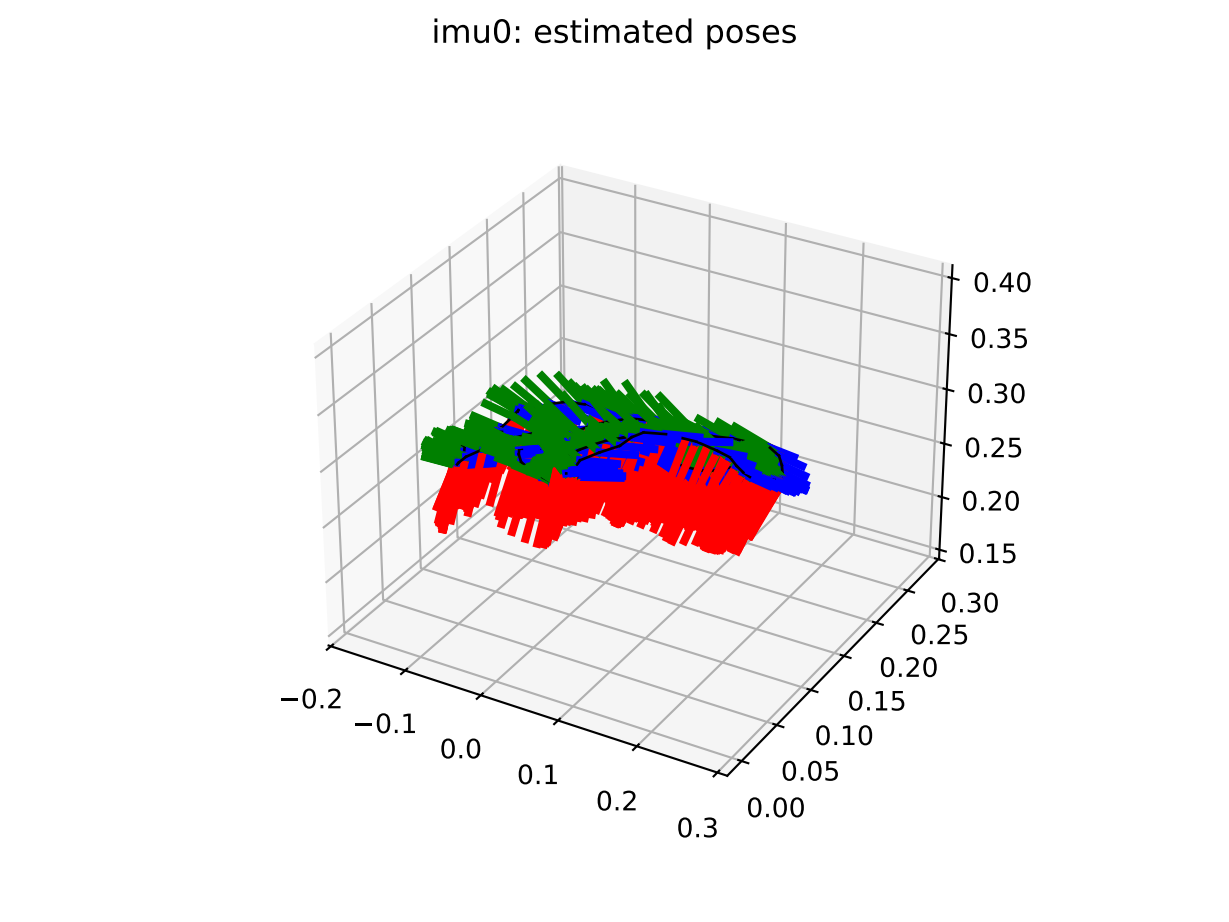
  Camera model: pinhole
  Focal length: [652.879962242561, 652.6913800309958]
  Principal point: [402.7459765146756, 293.51689967360625]
  Distortion model: radtan
  Distortion coefficients: [0.19004630188120233, -0.3566296810567926, -0.001643546078163314,
0.0007208195618292788]
  Type: aprilgrid
  Tags:
    Rows: 6
    Cols: 6
    Size: 0.017733333333333334 [m]
    Spacing 0.00532 [m]

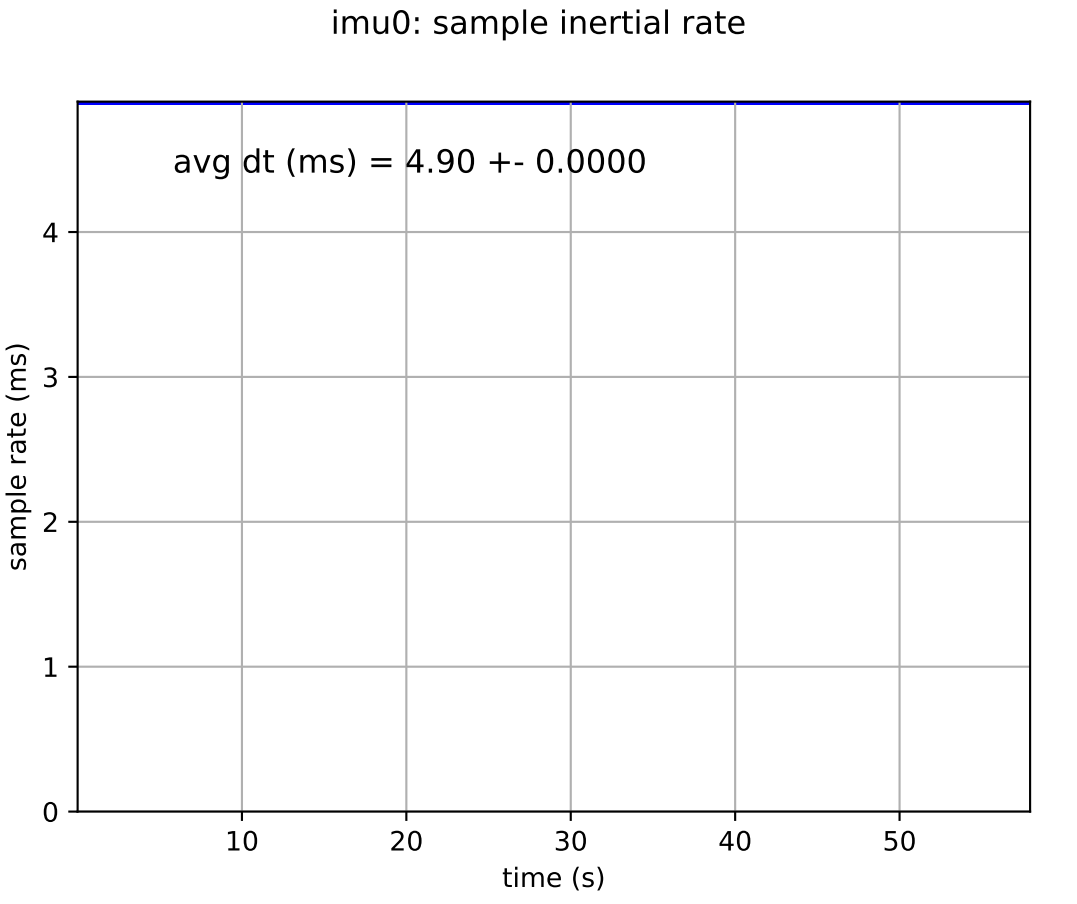
IMU configuration
==========

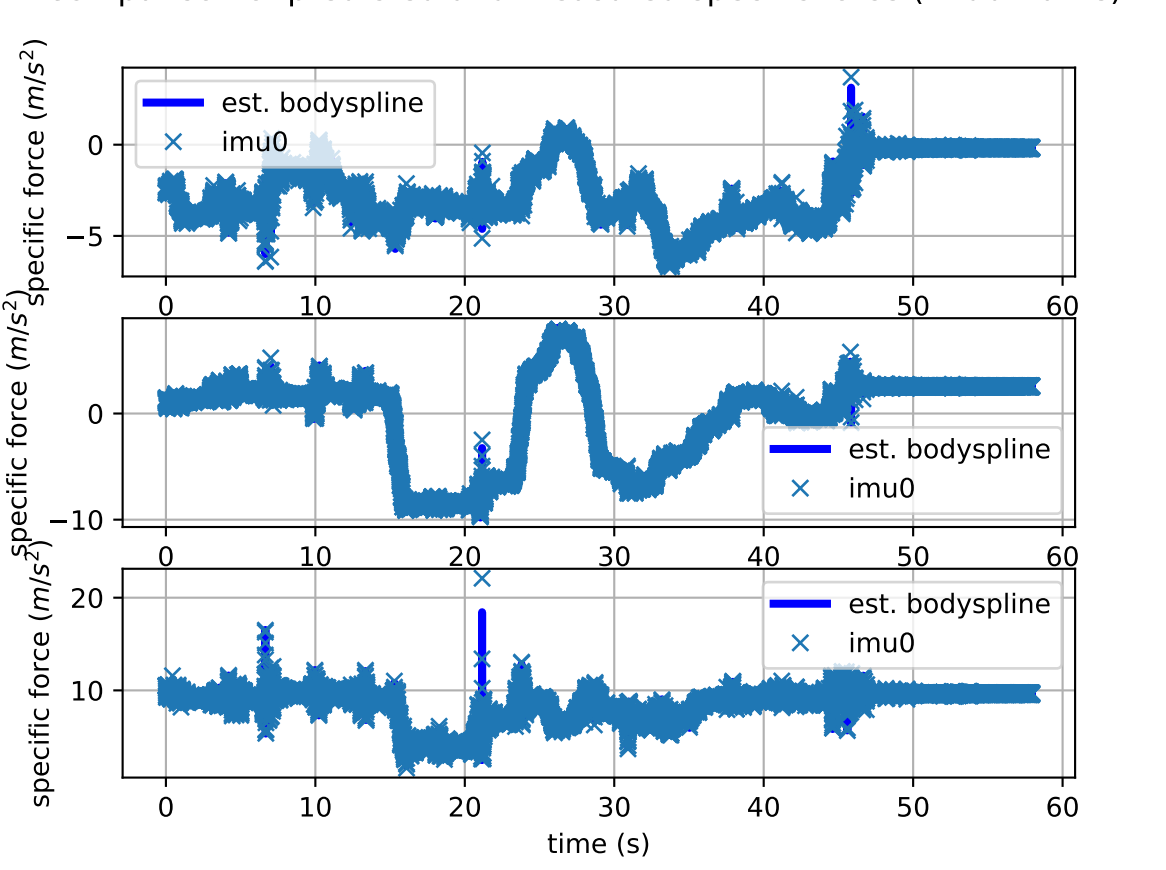
IMU0:

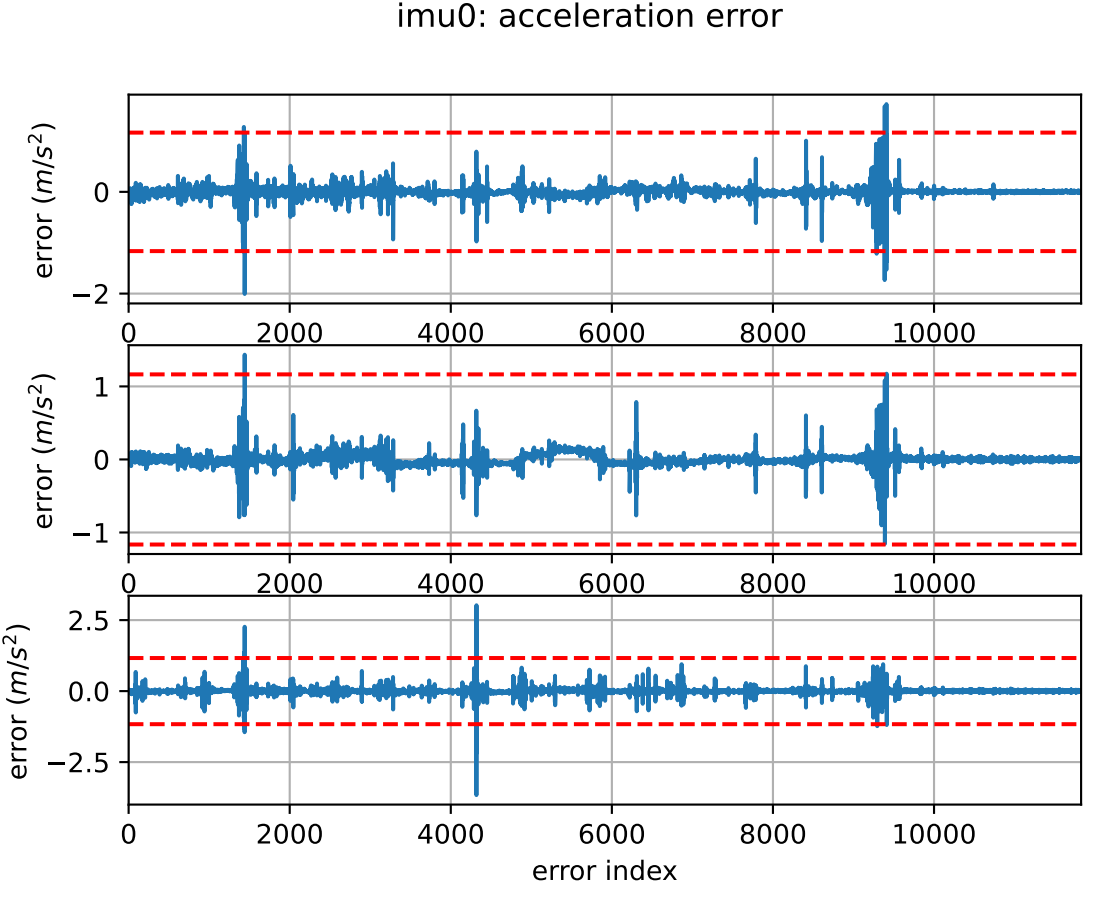
  Model: calibrated
  Update rate: 204.090470950253
  Accelerometer:
    Noise density: 0.027176766288825538
    Noise density (discrete): 0.38824792519257595
    Random walk: 0.0067023621585376265
  Gyroscope:
    Noise density: 0.002763830369432253
    Noise density (discrete): 0.039484145947029654
    Random walk: 0.00014770260860981916

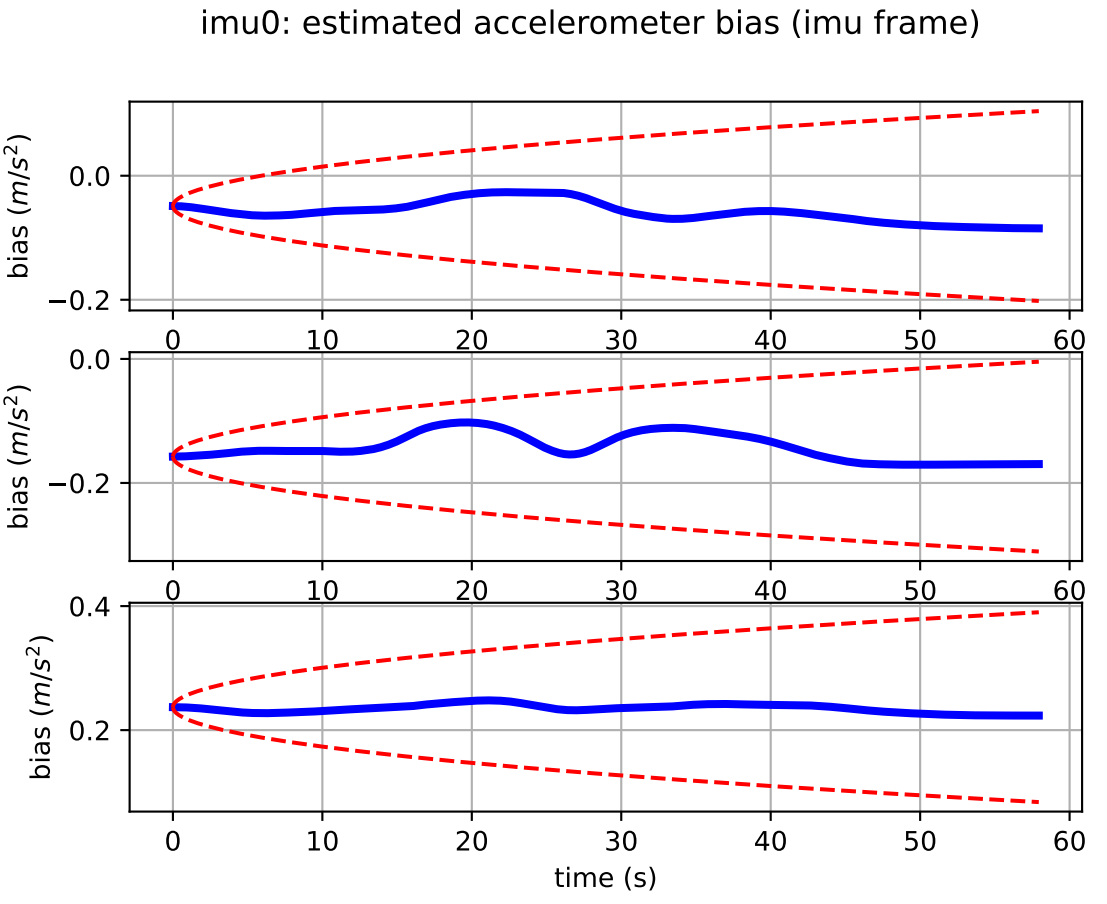
T_ib (imu0 to imu0)
  [[1. 0. 0. 0.]
   [0. 1. 0. 0.]
   [0. 0. 1. 0.]
   [0. 0. 0. 1.]]
time offset with respect to IMU0: 0.0 [s]

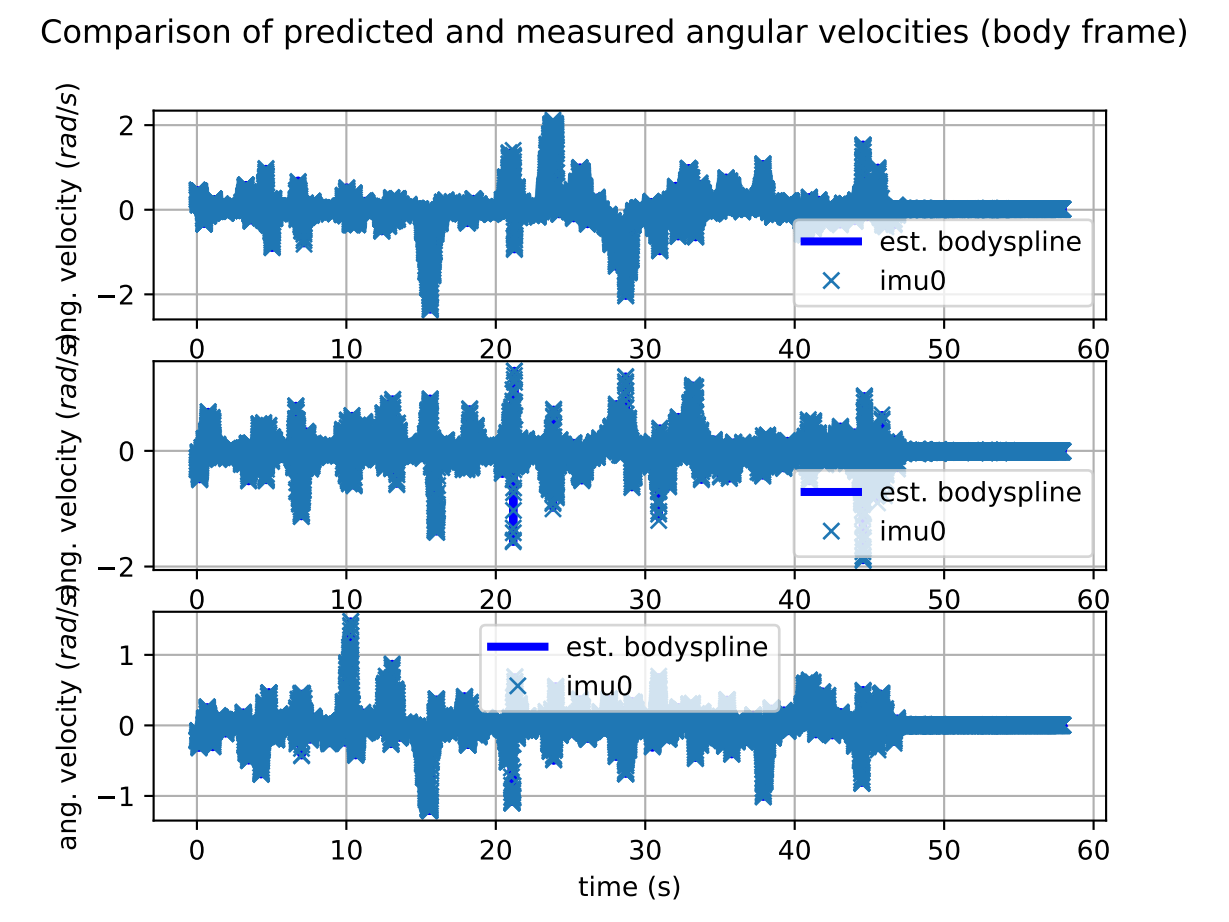

imu0: estimated poses
0.40
0.35
0.30
0.25
0.20
0.15
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−0.2
−0.1
0.0
0.1
0.2
0.3

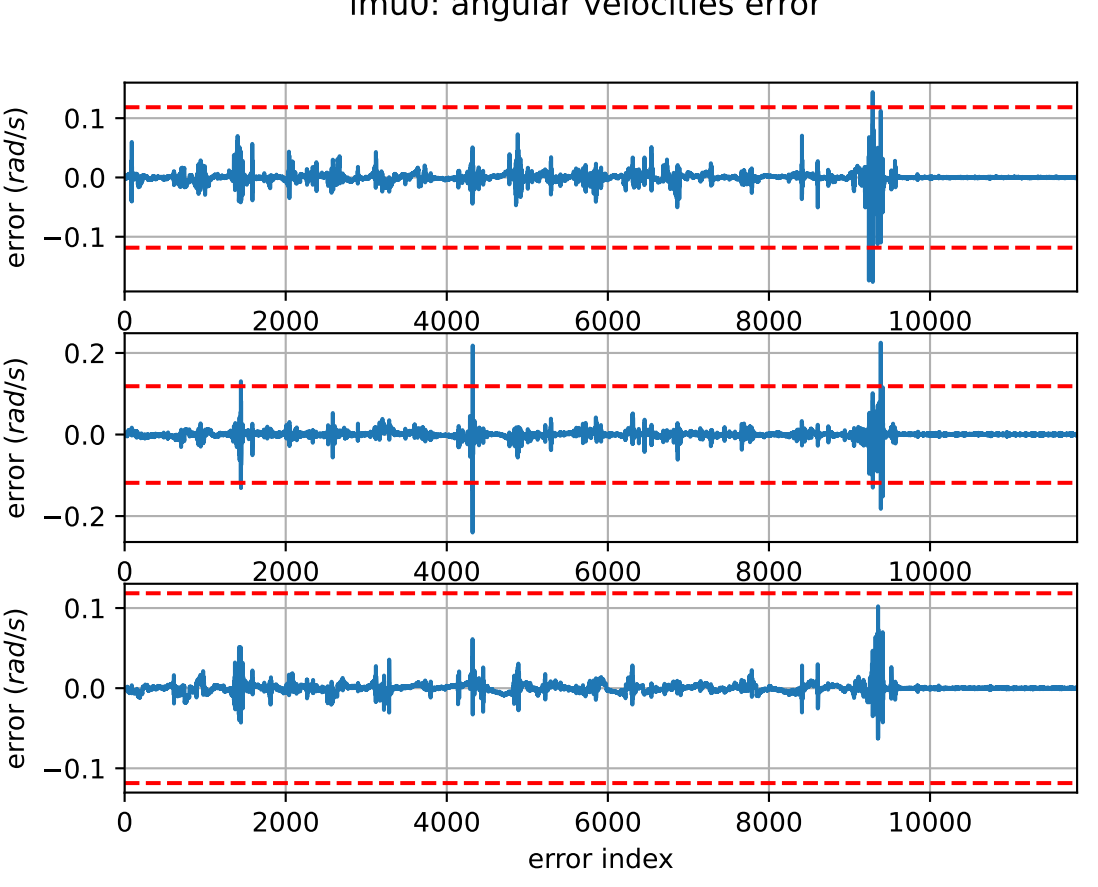

imu0: sample inertial rate
avg dt (ms) = 4.90 +- 0.0000
sample rate (ms)
0
1
2
3
4
10
20
30
40
50
time (s)

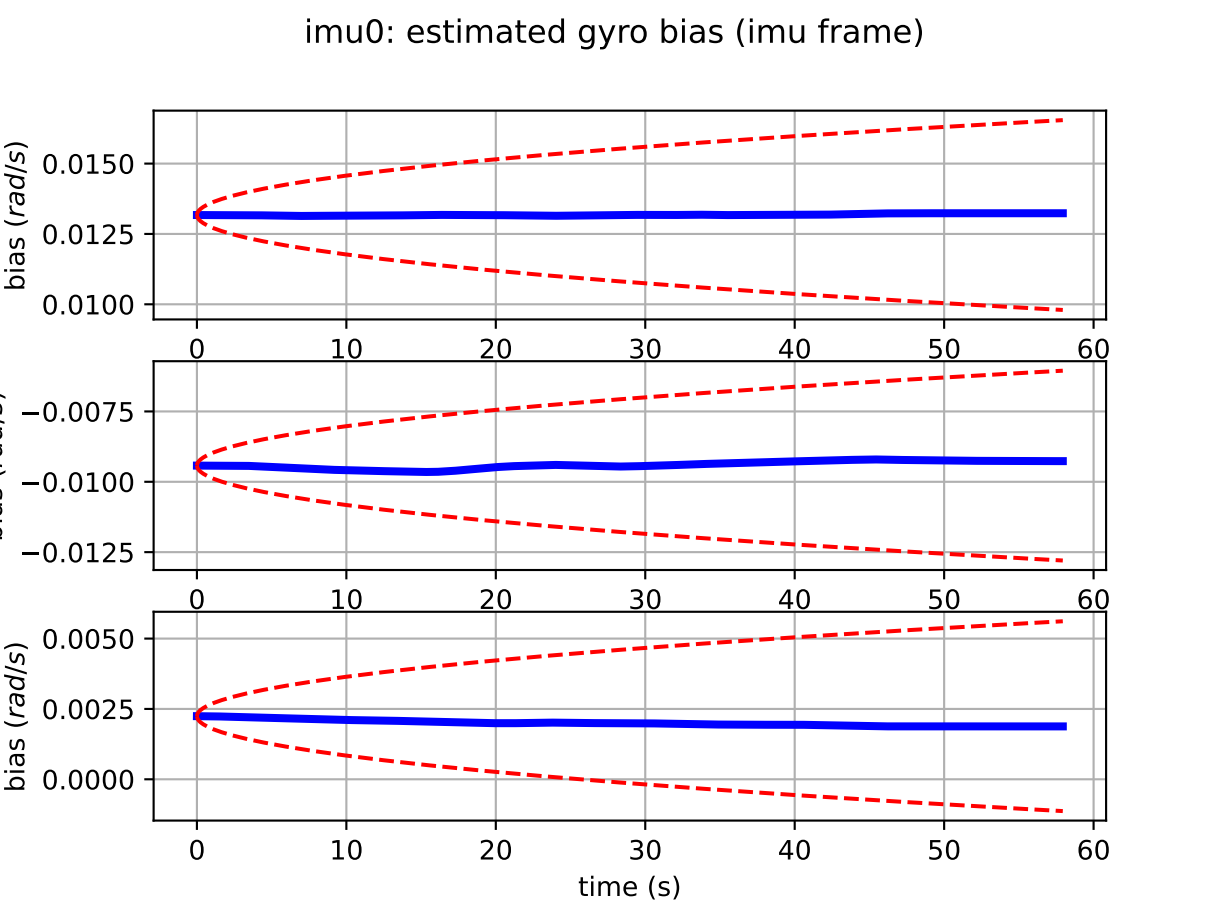

Comparison of predicted and measured specific force (imu0 frame)
specific force (m/s^2)
est. bodyspline
imu0
0
−5
0
10
20
30
40
50
60
specific force (m/s^2)
est. bodyspline
imu0
0
−10
specific force (m/s^2)
est. bodyspline
imu0
20
10
time (s)

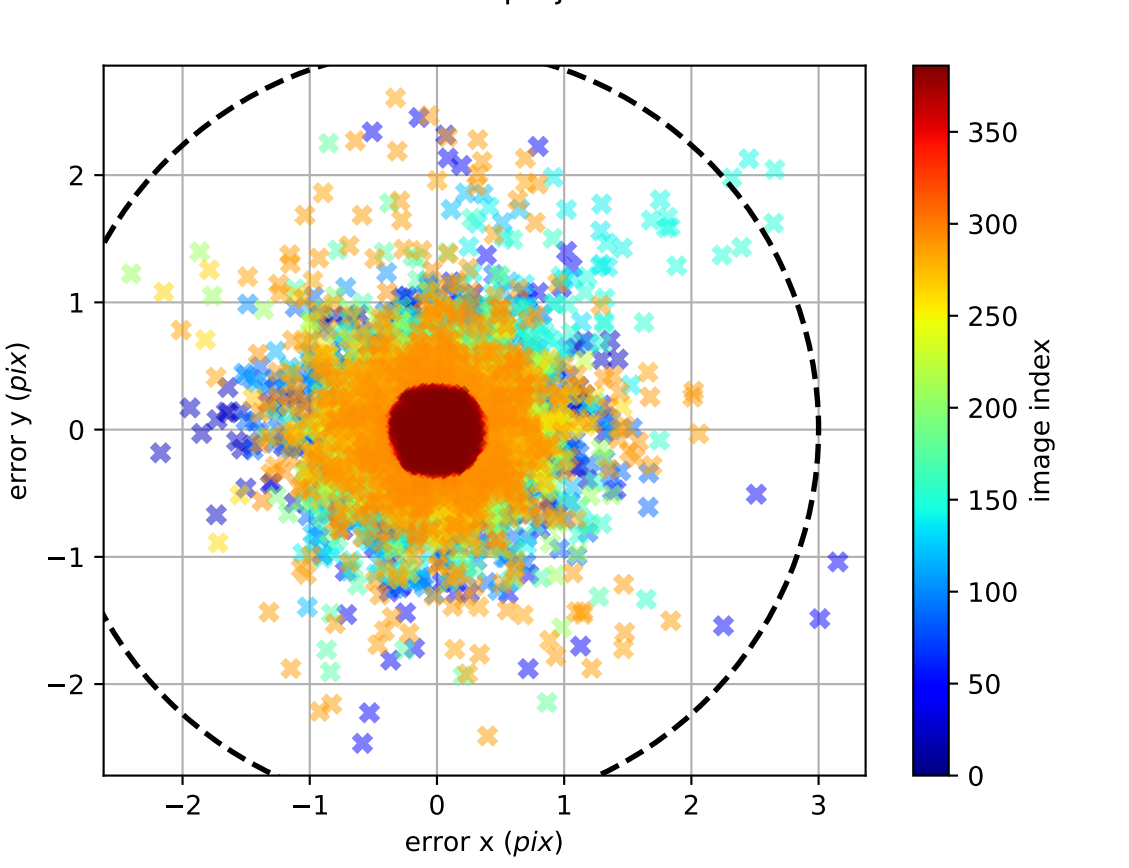

imu0: acceleration error
error (m/s^2)
0
−2
0
2000
4000
6000
8000
10000
error (m/s^2)
1
0
−1
0
2000
4000
6000
8000
10000
error (m/s^2)
2.5
0.0
−2.5
0
2000
4000
6000
8000
10000
error index


imu0: estimated accelerometer bias (imu frame)
bias (m/s^2)
0.0
−0.2
0
10
20
30
40
50
60
bias (m/s^2)
0.0
−0.2
bias (m/s^2)
0.4
0.2
time (s)


Comparison of predicted and measured angular velocities (body frame)
ang. velocity (rad/s)
ang. velocity (rad/s)
ang. velocity (rad/s)
time (s)
est. bodyspline
imu0
est. bodyspline
imu0
est. bodyspline
imu0


imu0: angular velocities error
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error (rad/s)
0.2
0.0
−0.2
0
2000
4000
6000
8000
10000
error (rad/s)
0.1
0.0
−0.1
0
2000
4000
6000
8000
10000
error index


imu0: estimated gyro bias (imu frame)
bias (rad/s)
0.0150
0.0125
0.0100
−0.0075
−0.0100
−0.0125
bias (rad/s)
0.0050
0.0025
0.0000
0
10
20
30
40
50
60
time (s)


cam0: reprojection errors
error x (pix)
error y (pix)
image index
−2
−1
0
1
2
3
2
1
0
−1
−2
350
300
250
200
150
100
50
0